

1. Suppose $E(X) = \mu$ and $\text{Var}(X) = \sigma^2$. Using a first-order Taylor series approximation about μ , find the approximate mean and variance of
 - (a) e^x
 - (b) $1/X$ (assuming $\mu \neq 0$)
 - (c) $\ln(X)$ (assuming $X > 0$)
2. Let X be uniform on $(0,1)$, let Y_n be exponential with mean $1/n$, and let $X_n = X + Y_n$. Show that X_n converges to X in mean square (second-mean L^2).
3. Consider a sequence of independent random variables indexed by $n \in \{1, 2, 3, \dots\}$, where

$$X_n = \begin{cases} 0 & \text{with probability } \frac{n-1}{n} \\ \sqrt{n} & \text{with probability } \frac{1}{n} \end{cases}.$$

To what does X_n converge (if it converges at all)? Prove or disprove that X_n converges to that quantity in the following modes.

- (a) In Distribution
 - (b) In Probability
 - (c) In First Mean (or L^1)
 - (d) Almost Surely
4. CB 5.18 (parts c–e) *Refer to HW 10 for 5.18 (a–b). Note that part (d) should say “Use the results of parts (b) and (c) to argue that...”*.
 5. CB 5.32
 6. Let $X_1, X_2, \dots$ be iid and $X_{(n)}$ be the maximum order statistic.
 - (a) If $X_i \sim \text{exponential}(1)$, find the limiting distribution of $X_{(n)} - \log(n)$.
 - (b) If the cdf of X_i is $F(x) = 1 - x^{-2}$ for $x > 1$ and 0 otherwise, find the limiting distribution of $Y_n = n^{-1/2}X_{(n)}$.
 7. Let $X_1, X_2, \dots$ be iid from a $\text{Beta}(1, \beta)$ distribution and let $X_{(n)}$ be the maximum order statistic. Find the limiting distribution of $n^{1/\beta}(1 - X_{(n)})$.
 8. CB 5.35
 9. CB 5.36 (On part b, consider the MGF of Y . Then, use the MGF of Y to get the MGF of Z . In this MGF, you should have a term like $+2\theta \frac{t}{\sqrt{8\theta-2t}}$ in the exponent. For this term, use the Taylor series expansion about 0 of $\frac{t}{\sqrt{8\theta-2t}}$. This will cancel out some terms but create many more. However, because we are considering $\lim_{\theta \rightarrow \infty} M_Z(t)$, almost all of these terms will go to 0. Good luck!)
 10. CB 5.39(part b)
 11. CB 5.44
 12. Let a be a real-valued constant and $X_1, X_2, \dots$ a sequence of random variables.
 - (a) Sketch the CDF of a . What does $X_n \xrightarrow{d} a$ mean?

- (b) Show that if $X_n \xrightarrow{d} a$ then $X_n \xrightarrow{p} a$. (That is, prove theorem 5.5.13.)
13. The homework consists of 40 independent problems. On the average, it takes 5 minutes to solve a problem, with a standard deviation of 2 minutes.
- (a) Find the approximate probability that the average homework problem takes more than 5 minutes 20 seconds.
- (b) Find the approximate probability that the homework will be completed in less than 3 hours.
14. An elevator fails if the total weight exceeds 2000 pounds. Suppose weights have a gamma distribution with mean 160 and sd 30. If 12 people get on the elevator, what is the approximate probability that it fails?
15. A keyword search program lists the files that contain a given keyword. If it runs through 200 files, and each file contains the keyword with probability 0.36, independently of other files, compute
- (a) the approximate probability that
- more than 70 files
 - less than 70 files
 - exactly 70 files will be listed
- (b) the approximate distribution of the odds that an average file contains the keyword.
16. Assume that maximum daily ozone level (in ppb) in January in Provo are distributed as $X_i \stackrel{iid}{\sim} \text{Gamma}(5, 10)$, using the parameterization in CB. Suppose that some quantified respiratory health risk $r = \exp(\bar{X})$; i.e., the exponent of the mean of the 31 daily maxima. What is the approximate distribution of r ?

Exponential Families

17. Show that each of the following families is an exponential family.
- Normal family with parameter μ known.
 - Normal family with parameter σ known.
 - Gamma family with parameter α known.
 - Gamma family with parameter β known.
 - Gamma family with both α and β unknown.
 - Beta family with parameter α known.
 - Beta family with parameter β known.
 - Poisson family.
18. Show that each of the following families is an exponential family.
- Negative Binomial family with r known, $0 < p < 1$.

- (b) Pareto with α known.
 - (c) Beta family with both α and β unknown.
19. For each of the following families, write the pdf in the canonical (natural) parameterization of the exponential family (equation 3.4.7), and use the identities

$$E(t_j(X)) = -\frac{\partial}{\partial \eta_j} \log c^*(\eta) \qquad \text{Var}(t_j(X)) = -\frac{\partial^2}{\partial \eta_j^2} \log c^*(\eta)$$

to find the **mean** and the **variance** of X .

- (a) gamma(α, β)
 - (b) Poisson(λ)
 - (c) Binomial(n, p)
20. CB 3.33. For a and b, part (ii), you can use the parameters of the Normal parameters (mean and variance) or the parameterization given by the exponential family. For c and d, part (ii), use the parameterization given by the exponential family.